JACK MURPHY

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EDUCATION

University of California, Los Angeles — **Los Angeles, CA**
Bachelor of Science in Mechanical Engineering, Emphasis in Electrical Engineering — *Expected Mar 2028*

- ***Cumulative GPA****:* 3.64/4.00
- ***Club Involvement***: Rocket Project @ UCLA
- ***Coursework***: Statics & Strengths of Materials, Dynamics of Particles and Rigid Bodies, Elementary & Intermediate Fluid Mechanics, Thermodynamics, Transport Phenomena, Electrical and Electronic Circuits, Introduction to Computer Science I & II

EXPERIENCE

Oriole Networks — **Palo Alto, CA**
Mechanical Engineering Intern — *May 2026 - Present*

- Designed and rendered extensive CAD models using SolidWorks and AI assisted Python scripting in FreeCAD
- Homogenized GPU chip aspects to vastly simplify meshing and modeled thin anisotropic materials using the shell method
- *In Progress:* Developing thermal simulation to assess temperatures and cooling requirements for GPU performance cards

Rondo Energy — **Alameda, CA**
Test Engineering Intern — *Jun 2025 – Sep 2025*

- Performed over 300 calorimetry tests and analyzed thermocouple (TC) data to calculate specific heat capacity (C_p) values
- Organized thermal cycling tester and TC measurement locations to test brick reliability under steep thermal gradients
- Compared brick stress behavior with thermal simulation predictions and presented results to Test, Materials, and Thermal teams

PROJECTS & RESEARCH

Carbon Hydrofoil Mast Construction — **San Mateo, CA**
Independent Engineering Project — *Apr 2026 – Jul 2026*

- Designed, hand built, and tested a high modulus full carbon hydrofoil mast following a NACA 0012 curve, 120 mm chord
- Programmed 3-axis CNC toolpaths in Fusion 360 to make mold halves for the mast, selected cutting tools, and speeds/feeds
- Wet-laid and pressed unidirectional and ±45º biaxial carbon plies (for torsional and linear rigidity) using 2 part 1:1 epoxy system
- Performed FEA in SolidWorks Simulation and physical static load tests under ~750 N (bodyweight) to test mast strength
  - Results: 4.6 MPa maximum stress, 0.0143 mm displacement under 2500 N axial load, passed 750 N static test

Computational Fluid Dynamics — **Los Angeles, CA**
Independent Fluid Mechanics Project — *Apr 2025 – Jun 2025*

- Conducted basic CFD analysis of F1 front wing using SolidWorks Flow Simulation, Python's NumPy and Matplotlib libraries
- Used flow trajectories to observe wing traits such as the stagnation point, boundary layer, eddy behavior and velocity deltas
- Gathered data using velocity point trajectories, found differentials of up to 90m/s, and calculated downforce using Bernoulli's eq.
- Experimented with advanced mesh refinement and assessed limitations of Bernoulli's eq. versus a Navier Stokes CFD approach

Statistical Entropy — **Los Angeles, CA**
University Research — *Jan 2025 – May 2025*

- Assisted Professor David Saltzberg in statistical mechanics research centered around Boltzmann's entropy formula
- Analyzed several statistical entropy derivations and examined extensive vs. intensive thermomechanical parameters

Mechanical and Electrical Laboratory — **Los Angeles, CA**
UCLA Physics Laboratory — *Sep 2024 – Mar 2025*

- Programmed an ESP32, ultrasonic sensor, and accelerometer to collect voltage, position, and acceleration data over time
- Graphed theoretical & experimental best-fit voltage curves and calculated time constant values for varying RC circuits in Python
- Used least squares method to calculate damping factors, theoretical frequencies, and fit Lorentzian curves to RLC circuit data

EXTRACURRICULAR

Competitive Sailor — **San Francisco, CA**
Multiple Boat Classes — *Lifelong Involvement*

- Coached under two-time Olympian and sports psychologist Udi Gal, training ~20hrs/week throughout high school
- *Notable finishes:* 2023 North Americans (5th), 2021 West Coast US Open (1st), 2025 Rolex Big Boat Series (12th)

ADDITIONAL

Skills: NREMT Certified, Java, Python, C++, MATLAB, SEM, SolidWorks, ESP 32, Fusion 360 CAM, CNC, Composite Layup
Interests: Computational Fluid Dynamics, Cars, Dinghy Sailing, Foiling, Kitesurfing, Mathematics, Mountain Biking, Skiing